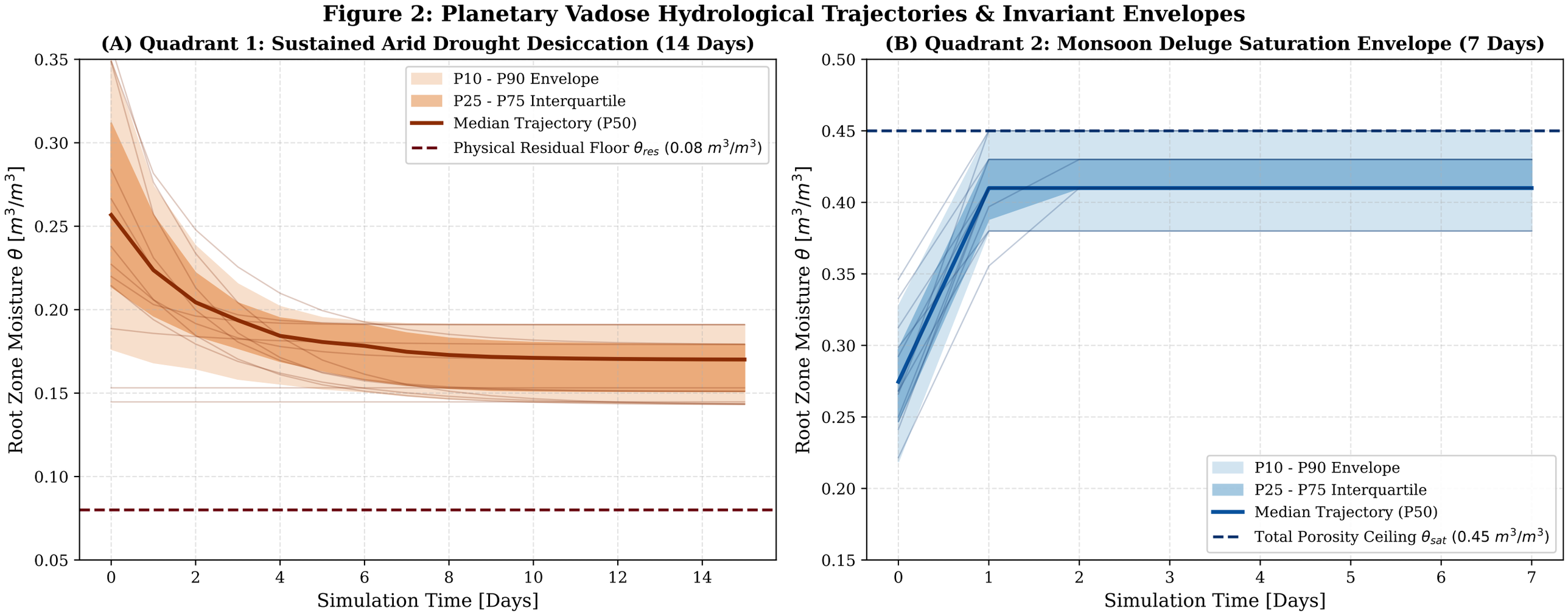
Figure 2: Planetary Vadose Hydrological Trajectories & Invariant Envelopes
(A) Quadrant 1: Sustained Arid Drought Desiccation (14 Days)
(B) Quadrant 2: Monsoon Deluge Saturation Envelope (7 Days)
Root Zone Moisture θ [m³/m³]
Simulation Time [Days]
P10 - P90 Envelope
P25 - P75 Interquartile
Median Trajectory (P50)
Physical Residual Floor θres (0.08 m³/m³)
Total Porosity Ceiling θsat (0.45 m³/m³)